THE RELATIVE FROBENIUS IS A UNIVERSAL HOMEOMORPHISM, INTEGRAL, AND PURELY INSEPARABLE ON RESIDUE FIELDS

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is a compact composition argument; every invoked lemma is named in the span and the factorization identity is displayed.

Allowed standard prerequisites (editorial).

- Absolute Frobenius is a universal homeomorphism. chapter lemma lemma-frobenius (not delivered as its own card). Gives the two universal homeomorphisms F_X and h.
- Permanence of universal homeomorphisms under composition; universal homeomorphisms are integral and universally injective with purely inseparable residue extensions. Morphisms, Lemmas morphisms-lemma-permanence-universal-homeomorphism, morphisms-lemma-universal-homeomorphism, morphisms-lemma-universally-injective. Deliver the three conclusions.

Source reference. Original lemma 0CC8: sources/stacks-chow/tags.txt, source lines 15498–15498. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0H2M: sources/stacks-chow/tags.txt, source lines 21287–21287. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 04DF: sources/stacks-chow/tags.txt, source lines 5725–5725. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01S4: sources/stacks-chow/tags.txt, source lines 2325–2325. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $p > 0$ be a prime number. Let S be a scheme in characteristic p. Let X be a scheme over S. Then the relative frobenius $F_{X/S} : X \to X^{(p)}$ is a universal homeomorphism, is integral, and induces purely inseparable residue field extensions.*

Proof. By Lemma 0CC8 the morphisms $F_X : X \to X$ and the base change $h : X^{(p)} \to X$ of F_S are universal homeomorphisms. Since $h \circ F_{X/S} = F_X$ we conclude that $F_{X/S}$ is a universal homeomorphism (Morphisms, Lemma 0H2M). By Morphisms, Lemmas 04DF and 01S4 we conclude that $F_{X/S}$ has the other properties as well. □

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.